

Patras, 28/07/2026

# **MOBIUS REPORT-MANUAL**

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## 2 INTRODUCTION

This manual details the operation and architecture of the developed 700V and 800V variants of the Unmanned Aerial Vehicle Power Station (UAVoPoS) for tethered drones. The system is engineered to minimize dependence on the onboard battery stack, enabling near-infinite flight times dictated primarily by the propulsion profile rather than battery capacity.

### Key Design Features:

- **Continuous Flight Duration:** Primary power is supplied via a ground-based tether.
- **Hybrid Power & Peak Shaving:** The miniaturized onboard battery stack is retained to assist during high-load acceleration phases and manage peak current demands.
- **Redundancy & Safety:** The onboard battery serves as an emergency fail-safe power source if tether power is interrupted.
- **High-Efficiency Transfer:** The tether utilizes a High Voltage link (400 to 800V) to significantly reduce cable losses over long distances.

### 2.1 POWER SYSTEM ARCHITECTURE

The complete power system utilizes a hybrid architecture to safely and efficiently route power to the UAV. It consists of two primary hardware stations connected via the tethered cable.

#### 2.1.1 Ground Power Station (GoPoS)

The GoPoS is the mobile power foundation for the system providing power to the UAV through the tether. Its main features are:

- **Input:** Operates on a standard single-phase power supply, ensuring compatibility in mobile or remote deployments where three-phase power is unavailable.
- **Output Control:** Adjusts the origin voltage ( $V_{GoPoS}$ ) fed into the bottom of the tether.
- **Telemetry Integration:** Features a communication interface to sync with UAV measurements and telemetry.

#### 2.1.2 The Power Tether

- **Function:** Serves as the physical and electrical link between the GoPoS and UAVoPoS.
- **Specifications:** Custom-designed to be lightweight and low-loss.

#### 2.1.3 UAV Power Station (UAVoPoS)

The UAVoPoS is the onboard hardware responsible for regulating tether power and integrating it with the drone's Electronic Speed Controllers (ESCs).

- **Voltage Step-Down:** A high step-down DC-DC converter reduces the 400–800V tether transmission down to the 20–60V operating range required by the Electronic Speed Controllers (ESCs) and motors.
- **Battery Integration (ORing Logic):** To safely interface the tether power with the onboard backup battery, the system utilizes a MOSFET controlled by an ideal diode controller.
- **Efficiency Note:** This "ideal diode" setup connects the battery to the output DC Bus through a single semiconductor, preventing reverse current flow while avoiding the massive thermal and power losses associated with standard power diodes under high-current loads.

The complete system architecture is presented in Fig. 1.

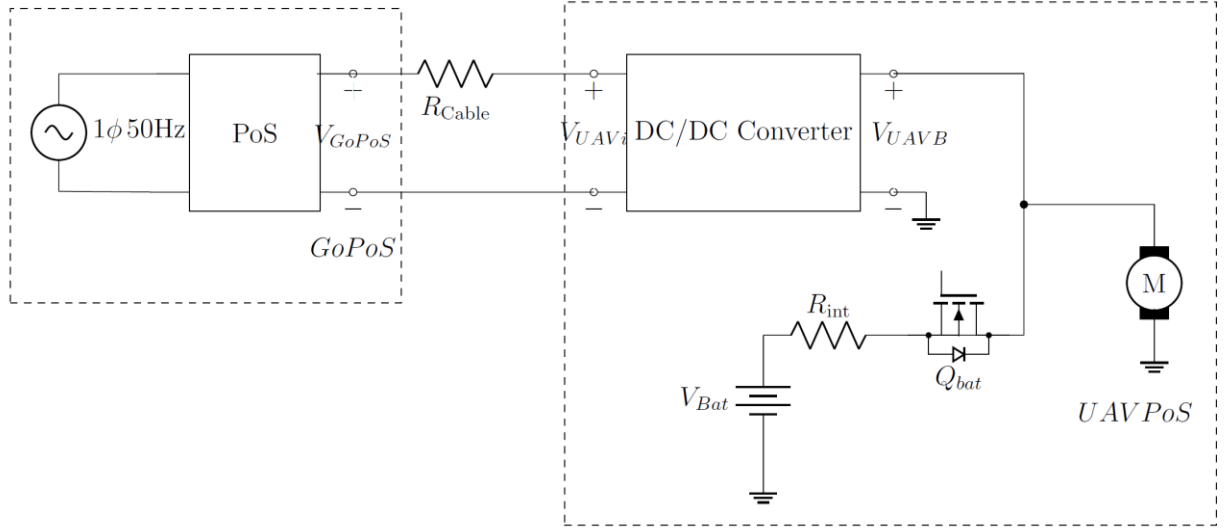


Figure 1: The complete power system architecture, including the GoPoS, the tether cable and the UAVoPoS

### 3 SYSTEM CONTROL & POWER MANAGEMENT

The primary objective of the power management system is to preserve the onboard battery stack's energy levels. The control algorithm is designed to ensure the tether acts as the primary power source at all times, with the onboard battery only intervening during power shortfalls or peak load demands.

To achieve this, the system manipulates a single control variable: the input voltage to the DC-DC converter ( $V_{UAVi}$ ). This voltage is actively managed from the ground via the programmable High Voltage Power Supply (HVPS) at the Ground Power Station (GoPoS).

#### 3.1 STANDARD OPERATING MODES

The power distribution system toggles between two distinct operational states depending on flight demands:

- Mode 1 (DC-DC Converter Exclusive Power): The standard operational state. The tether and step-down converter supply 100% of the power required by the load. The battery is in cutoff because of the reverse polarized diode.
- Mode 2 (Dual Source Constant Power): The high-load operational state. Both the tether and the onboard battery stack operate in parallel to meet the energy demands of the motors.

#### 3.2 CONTROL LOGIC & HYSTERESIS TRANSITION

The transition between modes is fully automated and governed by load demand as shown in Figs 2 and 3. The control unit at the GoPoS continuously monitors telemetry from the UAV—specifically voltages and currents ( $V_{UAVB}$ ,  $I_{o-c}$ ,  $I_{o-b}$ ).

- **Initialization (Mode 1):** The GoPoS output is set to a high baseline (e.g., 800V). This high voltage ensures the DC-DC converter output exceeds the battery stack voltage, keeping the battery's safety diode in reverse bias (preventing battery drain).
- **Approaching the Threshold:** As flight demands increase, the system monitors the power drawn from the DC-DC converter, denoted as  $P_o(n)$ . If  $P_o(n)$  rises compared to the previous sample, the GoPoS proactively lowers its output voltage, moving the system closer to a Mode 2 transition. Conversely, if power demand drops, the voltage is increased to stay safely in Mode 1.
- **Engaging Mode 2:** To prevent rapid, erratic switching between modes, the system utilizes a hysteresis window defined by boundaries at 1.65kW and 1.85kW. If load demand continues to surge and crosses the upper threshold, the system shifts into Mode 2.
- **Battery Contribution Management:** Once in Mode 2, the onboard battery voltage naturally exceeds the regulated tether voltage, forward-biasing the diode and allowing the battery to discharge into the system. As the total power requirement scales toward the 4kW system maximum, the GoPoS voltage is lowered further, forcing the battery to cover the remaining power deficit.

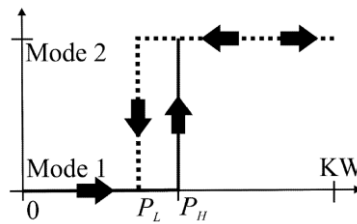


Figure 2: Power System Control Modes.

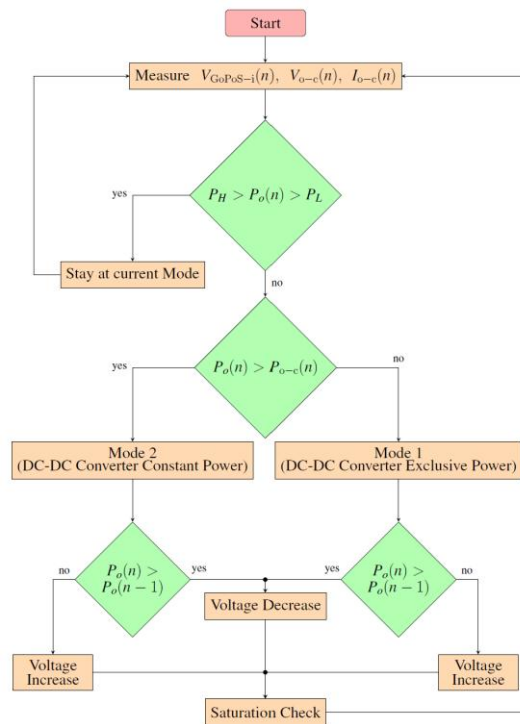


Figure 3: System's Control Flowchart

**Hardware Compatibility Warning:** The current control scheme requires the native voltage of the onboard battery to perfectly match the operating voltage of the Electronic Speed Controllers (ESCs). If future deployments utilize batteries with a higher or lower voltage rating than the ESCs, a secondary power converter must be installed between the battery and the output bus, and the control algorithm will require recalibration.

## 4 SELECTION OF BASIC SYSTEM COMPONENTS

In the following table, the required specifications for the developed UAVoPoS prototypes under the framework of the MOBIUS project are presented:

| Variant 1                 |          | Variant 2                 |          |
|---------------------------|----------|---------------------------|----------|
| Input Voltage             | 400-700V | Input Voltage             | 500-800V |
| Max. Output Power         | 1750W    | Max. Output Power         | 1750W    |
| Max. Output Current       | 40A      | Max. Output Current       | 35A      |
| Max. Battery Contribution | 2400W    | Max. Battery Contribution | 2400W    |
| Max. weight               | 1 kg     | Max. weight               | 1 kg     |
| Max. PCB Dimensions       | 16*9*7cm | Max. PCB Dimensions       | 16*9*7cm |

### 4.1 GoPoS

To minimize the voltage drop across the tether cable and to reduce its overall size, a high voltage level is selected for the transmission of power from the GoPoS to the UAVoPoS. The specified voltage range of 400 to 800V is considered ideal. The maximum requested power from the GoPoS (1750W) can be supplied by a single-phase power supply. During testing at UPAT, the ITECH it6018c-1500-40 power supply was used. However, the BK Precision PVS10005 power supply can be used as well.

### 4.2 TETHER CABLE

Any cable suitable for UAV tethering might be used. LINDEN has proven to be a reliable manufacturer of tether cables. If fiber optics communication is required, LINDEN-SPE-7155 may be used.

### 4.3 UAVoPoS

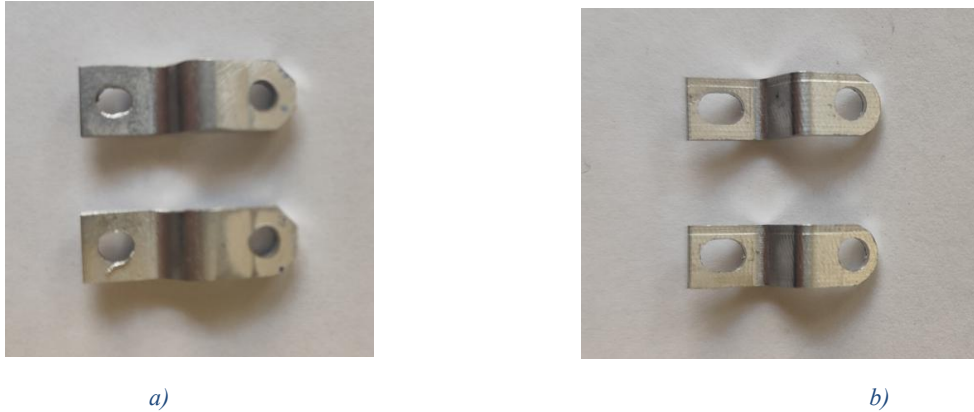
#### 4.3.1 DC-DC Converter

This is the station of interest for the MOBIUS project. First the DC-DC converter for the reduction of the voltage level from the 400-800V to the ESC voltage level is selected. To account for the two variants requested for the voltage input range the following two modules from Vicor are considered ideal:

1. Variant 1: BCM4414VG0F4440M02 – 400-700V input, 25-43.8V output, max. output current 40A
2. Variant 2: BCM4414VH0E5035T02 – 500-800V input, 31.3-50V output, max. output current 35A

Both modules are chassis mount so that increased modularity can be achieved and they both have an operating temperature range from -40°C to 100°C. Additionally, these modules have a built-in telemetry system that can be accessed via PMBus.

The interface of the chassis mounted Vicor modules with the PCB is performed through the designed busbars that are displayed in Fig. 4. The material used for their fabrication is aluminum 6061T6.



*Figure 4: a) Low Voltage and b) High Voltage Bus Bars*

To gain advantage of the full power of the modules, a heatsink must be mounted on top. For the developed prototypes, the official heatsink is selected (Heat Sinks for VIA 4414 Longitudinal Fins 22.86 mm tall). Additionally, a three 5V,0.24A fan forced convection was implemented. Two fans were blowing air towards the heatsink (low voltage side and upper side fans), while the third one (high voltage side) was removing the hot air away from the 3D printed case of the system.

#### 4.3.2 Battery Interface

The battery stack selection is left for the UAV designer to decide. Any battery such as Tattu Plus 22000mAh 6S1P Lipo Smart Battery Pack may be used, **provided that its voltage level is well within the output voltage range of the module (variant 1 or 2)**. The interface of the battery stack to the output DC Bus is performed by driving the IPB010N06NATMA1 MOSFET via the LM74700QDBVRQ1 ideal diode controller. A current sensor was added so that the operator can have constant information on battery contribution.

#### 4.3.3 High Current Connectors

As the design request was to aim for connectors that are as secure as possible, the QS8 part number proved to be an ideal candidate. These connectors are not PCB solderable by themselves, therefore a metal bracket was designed and manufactured to serve as the medium between the QS8 connector and the PCB, as shown in Fig. 5.



*Figure 5: Metal brackets for the interconnection of QS8 connectors with the PCB.*

**Attention:** These parts cannot be manufactured with ALUM6061-T6 as this material is not solderable and therefore it will be nearly impossible to establish a reliable connection between QS8 and the metal part. Instead, there are several available options based on the machinery capabilities:

- Brass (Alloy C360) – Easy to machine, highly solderable, 3 times heavier than aluminum
- Copper (Alloy C110) – Perfect electrical and thermal conductivity, suitable for high currents, highly solderable, more difficult to machine
- Aluminum (6061-T6) – You may keep the aluminum material as it is the most lightweight, but after manufacturing they need to be sent to a metal finishing shop to be plated. Two main options for plating:
  - Electroless Nickel/Immersion Tin
  - ENIG (higher cost)

**NOTE: After careful design of the system, all design specifications were reached. Especially, the total weight of the device is ~560g, which is nearly half of the required weight.**

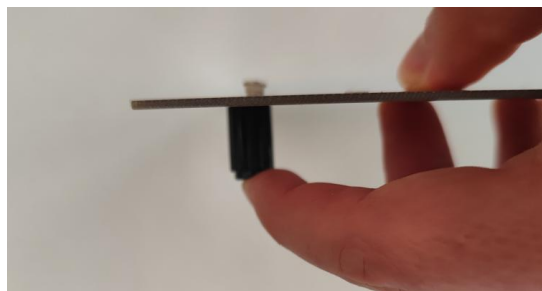
## 5 ASSEMBLY OF THE PROTOTYPES

1. As it is a slightly difficult process, first the M3 and M4 studs for the busbars are mounted in the PCB. In this way, if the process goes wrong, the PCB can be discarded without wasting the time and energy to solder the rest of the parts.
  - a. The mounting idea is that the stud must be pressed in the PCB without harming it. A vise can be used to apply pressure. Insert the stud into the PCB.



*Figure 6: Insert stud in the PCB hole.*

- b. Place a mold that exceeds the length of the stud around the stud. To do so, the mold needs to be hollow on the inside. The mold is used to fix the overall height of the board.



*Figure 7: Mold Placement.*



- c. Place the PCB with the mold in a vise.

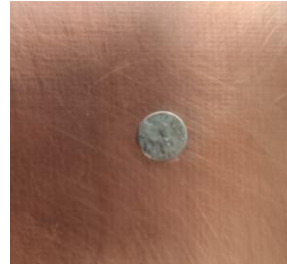


*Figure 8: Placement in vise.*

- d. Use the vise to apply pressure between the PCB and the top of the mold.  
e. Remove the mold. The stud should be placed as shown in Fig. 9.



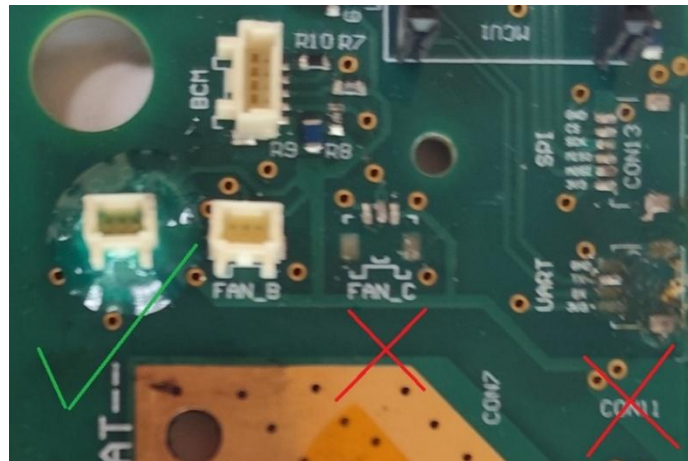
*a) Top view*



*b) Bottom view*

*Figure 9: Final result.*

2. All SMD and low-profile components are soldered in place.
3. The BCM communication, UART, SPI and the three fan connectors need to be secured with epoxy resin. Otherwise, there is a significant possibility that the connectors might be destroyed during insertion/extraction, taking away with them the pads from the PCB.



*Figure 10: Resin protected vs unprotected connectors.*



4. Apply thermal paste in the designated area over which the BCM module will be placed.



*Figure 11: Thermal paste application.*

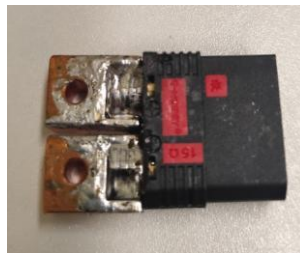
5. Unscrew the high and low voltage screws where the metal bus bars will be connected.
6. Place the BCM module in the designated area.
7. Prior to placing the metal bus bars, please place electrically conductive paste on the PCB pads as shown in in Fig. 12 to enhance connectivity between the bus bars and the PCB. For the prototype production, the MG chemicals 847 Carbon Conductive Assembly Paste was used. ATTENTION: Watch out so that the conductive paste doesn't leak and violates the required voltage clearance. The application of the conductive paste can be neglected from the high side and it must definitely be included in the low voltage-high current side. <https://mgchemicals.com/products/grease-for-electronics/electrically-conductive-grease/conductive-paste/>



*Figure 12: Place conductive paste on the bus bar pads.*

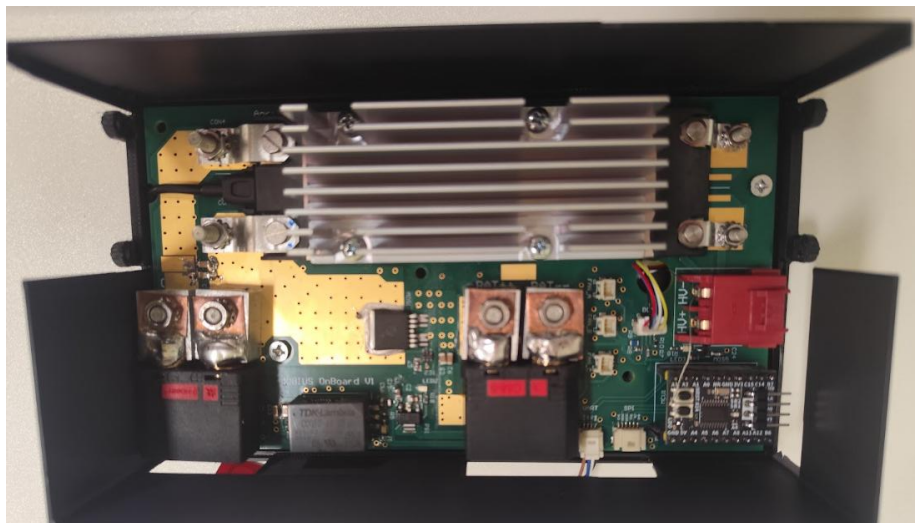
8. Place the high voltage bus bars. Screw them in about halfway as it goes. Use the screws provided by the manufacturer.
9. Place the low voltage bus bars. As the aluminum width is thick so that the bus bar can withstand the current, the screws provided by the manufacturer do not have adequate length. Consequently, third party screw must be used. For the prototypes, M3.5 screws with a length of 7.5 to 8mm were used. Screw them halfway as well.

10. Apply thermal paste at the top side of the Vicor module.
11. Place the heatsink on top.
12. Mount the heatsink, using M3 screws with kep nuts or nuts, flat washer and grower for a secure mounting. **Attention:** Watch out so that no voltage clearance is violated when the nuts/washers are placed. Screw half way through.
13. Screw tightly all screws so that the module, bus bars and the heatsink are secured. **Attention:** do not apply extreme force, otherwise the module might be damaged.
14. Solder the QS8 female connectors to the metal brackets. A mold should be used for the brackets to be aligned, so that they can properly placed on the pad surface of the PCB.



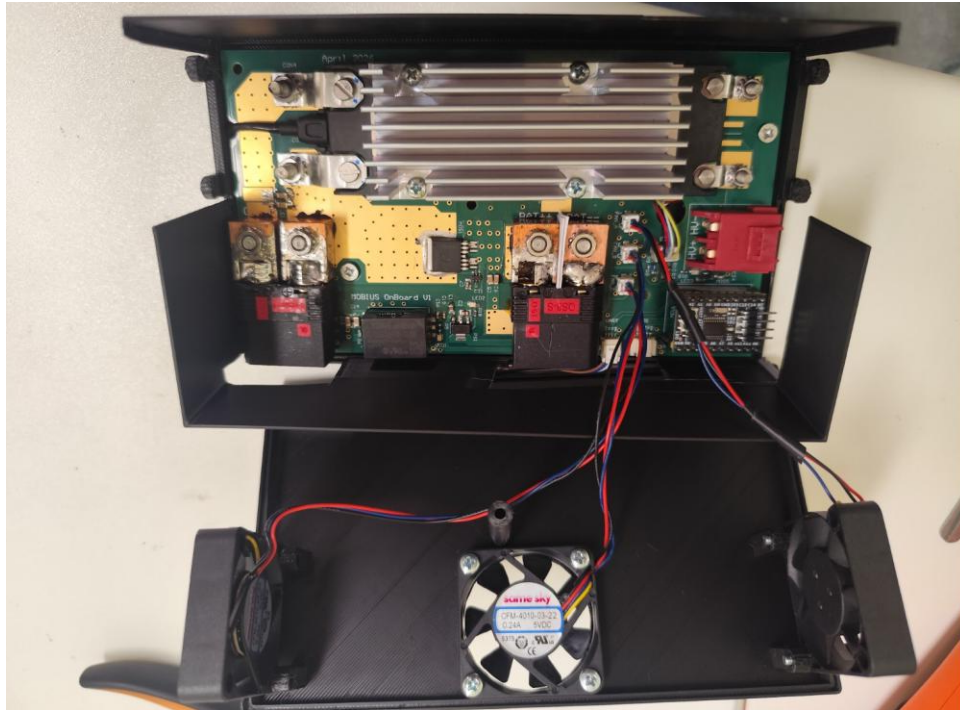
*Figure 13: QS8 connector soldered to bracket.*

15. Place them on the corresponding pads and screw the kep nuts to secure the position. **Attention:** You should verify that the two metal brackets of the connector are not producing any short with the PCB. Use the beeper function of a multimeter to verify that the two metal brackets are not connected. Do not rely on assessment by eye. You may need to apply Kapton tape at some parts of the PCB.
16. Assemble the signal cable of the BCM module with the proper connector crimps. Cut it shorter in length compared to the original to fit better in the case. You may consider to purchase factory assembled cables as their assembly is a cumbersome process.
17. Plug the signal cable to the BCM port and the PCB signal connector.
18. Mount the PCBA on the lower part of the 3D printed case.



*Figure 14: PCB mounted on the lower part of the case.*

19. Assemble the fan cables (Power, Ground, PWM) with proper connector crimps. Leave the cables of the fans lengthy enough to be able to close and open the upper part of the case with no issue.
20. Mount the fans on the 3D case. The one on top and the one in the low voltage side should blow into the case, while the one placed in the high voltage side should blow air out of it.



*Figure 15: Both parts of the case mounted with the respective components.*

21. Snap the two parts of the case together and secure the fans with extra screws.



*Figure 16: Assembly complete.*

## 6 TESTING THE PROTOTYPES

A procedure starting from simple operational tests and moving towards full functionality testing has been followed to validate the proper performance of the device.

### 6.1 PRELIMINARY TESTS

1. **Before conducting any test, please use the beeper function of a multimeter to verify that there is not short circuit among the positive and negative terminals of any connector.**
2. To verify that the low power electronics operate properly, a voltage between 20 and 30V is applied to the output DC Bus from an external power source. Power supply PS2 should turn on and output 5V. Then, power supply PS1 should output 3.3V. The relevant LEDs should be bright and so is the case for the small LED on the microcontroller board. This verifies that no short-circuit is present at the low power electronics after soldering.
3. Then, a high voltage (400V for variant 1 and 500V for variant 2) is applied to the input of the BCM module so that the powertrain and telemetry are turned on. No load is connected to the output. Dashboard.py or any other script that receives data from the UART port of the system is executed to verify that the MCU is correctly communicating with the module and is receiving data. Baudrate is 115200.
4. Next it should be safe for the prototype to handle power. An electronic load was connected to the output of the BCM module. NO battery was connected. The input voltage was varied through the whole range of each variant and so did the output current (through the electronic load).
5. Then, the battery was also connected to the system. The important aspect of this test is to verify that the MOSFET interfacing the battery is properly driven. To do this, the Vds voltage of the mosfet was measured with a multimeter, while the system was transitioning from the BCM only mode to the battery assistance mode. In battery assistance mode, the Vds should be as little as 20mV. It should NOT be around 0.7V as this would mean that the driving is not operating correctly and the mosfet is conducting through its body diode.
6. By conducting all previous tests, we made sure that the prototypes were fully operational and therefore we started testing them in more harsh conditions.

### 6.2 FULL TESTING OF THE DEVICES

#### 6.2.1 Load Variation Tests

First, different load schemes were applied to verify the operation of the system, provided that UPAT did not have any UAV to test it on. As will be shown, the devices demonstrate the following features:

- **Dynamic Response:** Both voltage variants automatically track and fulfill dynamic flight profiles, providing stable, continuous power during rapid load spikes and steady-state hovering.
- **High-Efficiency Baseline:** The architecture maintains exceptional efficiency because the backup battery interfaces through a single, non-switching MOSFET.
- **Cable Loss Factor:** Note that baseline efficiency metrics do not account for tether cable losses. Total system efficiency will scale inversely with the length of the deployed tether.



The system autonomously manages the transition between tether-exclusive power (Mode 1) and dual-source power (Mode 2) at approximately the 1.75kW threshold. Operators monitoring telemetry will observe the following behaviors during flight:

- **Entering High Demand (Mode 1 → Mode 2):** When a sudden load spike occurs (e.g., rapid ascent), the DC-DC module will initially supply the total required current for a micro-interval of roughly 25 milliseconds. Once the load surpasses the Ground Power Station (GoPoS) current limit, the ground station enters a current-limited state. This intentionally drops the input voltage at the UAVoPoS. As the voltage gap closes, the battery MOSFET engages, allowing the onboard battery to safely supplement the required power.
- **Returning to Standard Load (Mode 2 → Mode 1):** As flight demand normalizes and current draw decreases below the GoPoS limit, the ground station automatically reverts to constant voltage mode. The system voltage rises until the battery MOSFET naturally goes into cutoff, seamlessly returning the drone to Mode 1 with minimal voltage overshoot.

Two load profiles were tested: a) the Ramp and Step Variation and b) the Emulated UAV Acceleration Variation. The recorded behavior is presented below.

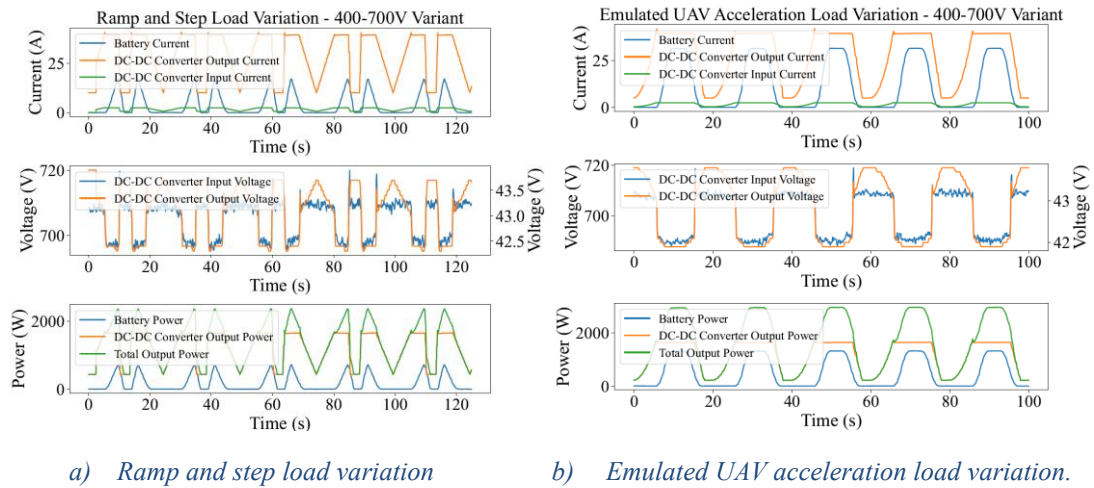
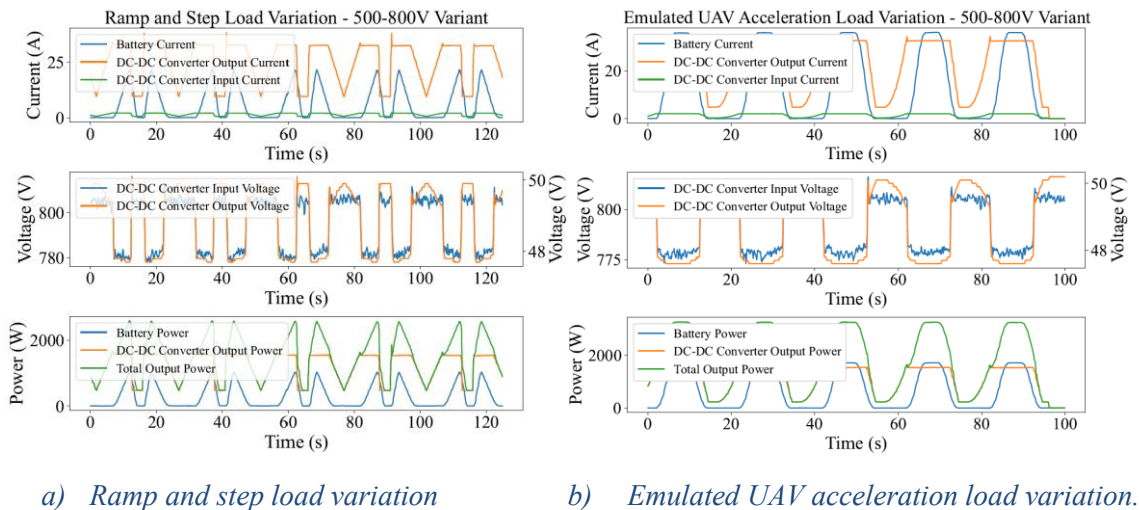
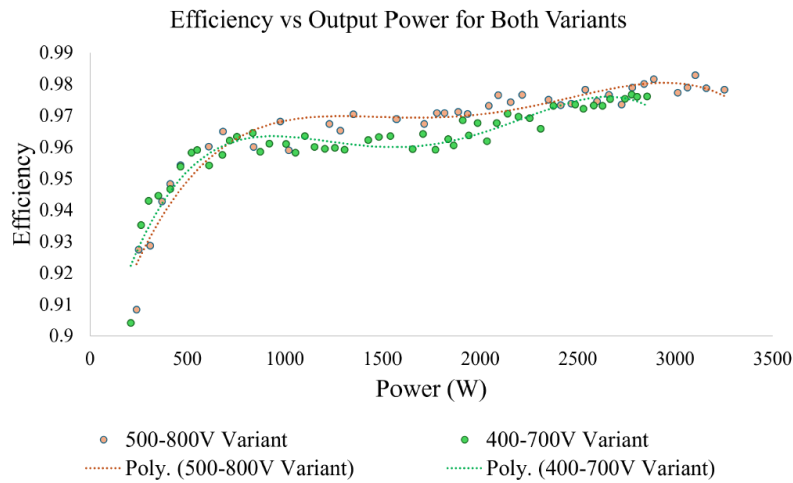


Figure 17: Load variation scenarios for the 400-700V variation.



*Figure 18: Load variation scenarios for the 500-800V variation.*

One of the major advantages of the proposed power architecture is its high efficiency, due to the interconnection of the battery stack with only one (non-switching) component and the selection of a low loss DC-DC converter. This is proven in this subsection, where the efficiency curves are extracted for both variants in Fig. 19. The transition from mode 1 to mode 2 after 1.75kW is also visible as the shape of the trendline is changing near this point. Tether cable losses are not included in the graph, as they are directly dependent on the length of the tether.



*Figure 19: Efficiency measurement curves for both variants under a wide output power range. Tether cable losses not included.*

### 6.2.2 Mode Transition

The seamless transition from Mode 1 (power offered solely by the DC-DC converter) to Mode 2 (battery stack assists the operation) and vice-versa is crucial for the overall operation and consequently it is studied here. For each system variant, the highest possible input voltage was used (700 and 800V accordingly). The current limit of the power supply at the GoPoS was set slightly lower to the maximum input current of each module.

Looking into Fig. 20a, the system operates in high power (30 A at the output) in Mode 1 in interval 1. Then, in interval 2, a 10A increase in the output load demand is performed. For a small period of time ( $\sim 25$ ms) in intervals 2 and 3, all the current is provided by the module as neither the overcurrent protection of the module nor the current limit protection of the power supply is yet tripped. Then, in interval 4, the current limit protection of the ground power supply is tripped, leading to a reduction at the input voltage of the UAVoPoS. Consequently, the output voltage is also reduced and so is the voltage difference among itself and the battery stack (yellow waveform). Finally, at the start of interval 5, the MOSFET is turned on, the battery stack contributes power to the output (Mode 2), while the input current of the module is set to the value of the current limit of the power supply.

The opposite transition (Mode 2 to Mode 1) is shown in Fig. 20b. In interval 1, the system operates in Mode 2 and both sources provide power. At the start of interval 2, a 10A decrease of the output load is performed and the current at the output and consequently at the input of the module decreases and falls below the current limit of the power supply. At the start of interval 3, the power supply starts to return to a constant voltage mode, increasing its voltage. Meanwhile, the body diode of the MOSFET goes into cutoff. Finally,

after a transition with a minimal overshoot due to the operation of the power supply, the system goes into its stable operation in Mode 1.

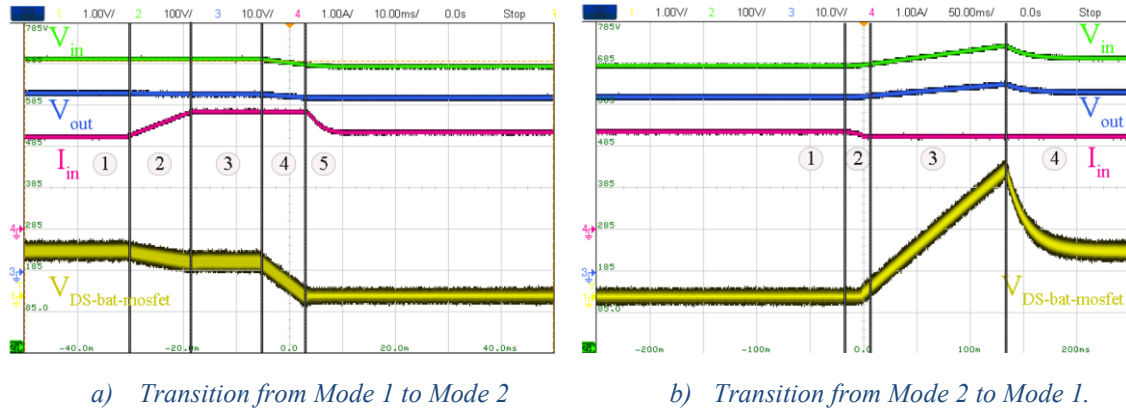


Figure 20: Mode transition validation for the 400-700V variant.

The exact same test for the mode transition is performed for this variant as well. Considering that the response of the system is identical, a detailed description of the transition is omitted and the corresponding scopes are presented in Fig. 21 for completeness.

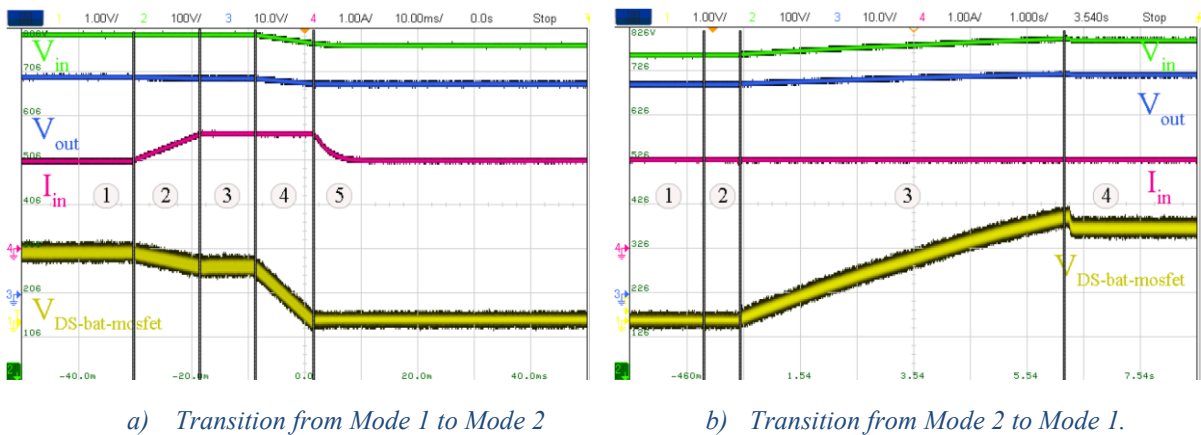


Figure 21: Mode transition validation for the 500-800V variant.

### 6.2.3 Thermal Testing

The design of the cooling system and the overall thermal behavior of the two basic components (DC-DC converter and Battery MOSFET) that handle the high power is presented here. The system is tested for the worst case scenario that can be implemented in our lab. The DC-DC converter is offering nearly full power (1500W) while the battery offers between 1750 and 2250W. To achieve that, the input of the UAVoPoS is initially set to the maximum (700V and 800V accordingly). However, as stated in the control system section, the voltage drops when the system transitions to mode 2, as the current limit of the power supply is activated. The cooling fans are operating as described in the system cooling subsection. To avoid disrupting the experiment as the UAVoPoS cases were closed, the thermal images presented below were taken after the completion of data logging, with the system still operating and having the top lid of the case opened.



The thermal testing results of this variant are presented in Fig. 22. Ambient temperature is  $23.5^{\circ}\text{C}$ . Initially the system operates in mode 1 and then it transitions to mode 2 as demanded power increases. The testing is concluded only when the system has reached a thermal equilibrium, meaning that its internal temperatures have not changed for at least one minute. In this case, the temperature of the DC-DC converter reaches  $63^{\circ}\text{C}$ , which refers to the internal temperature of the module, as the measurement is taken directly from its internal sensor. The temperature of the battery MOSFET reaches only  $27^{\circ}\text{C}$ , measured on its case, owing to its minimized conduction losses. From the thermal camera view, it can be verified that there are no visible thermal hot-spots.

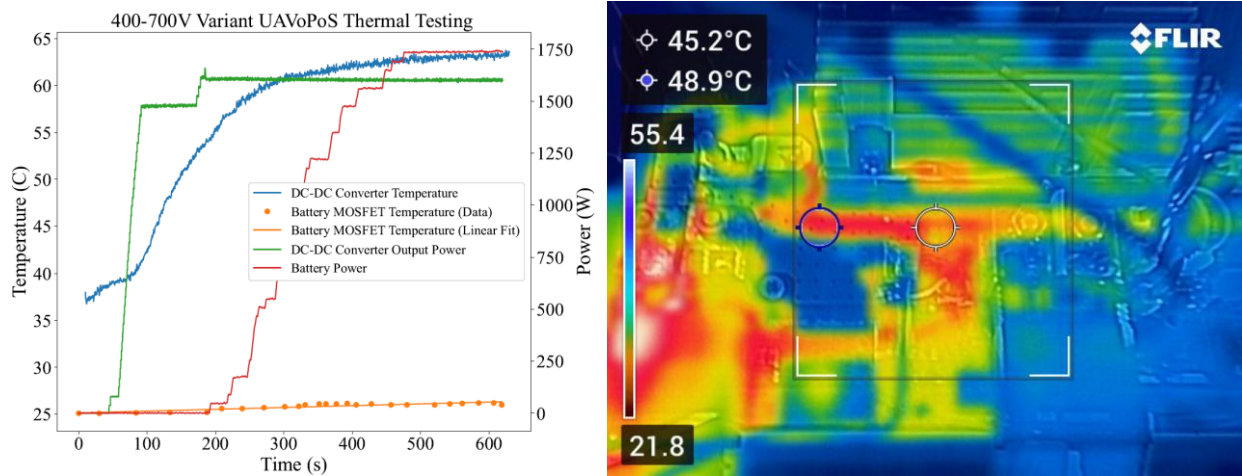


Figure 22: Thermal performance of the 400-700V variant UAVoPoS at full load.

The same testing procedure is followed in this variant as well and the corresponding results are shown in Fig. 23. Ambient temperature is  $25.7^{\circ}\text{C}$ . Here, the temperature of the DC-DC converter is reaching an equilibrium at  $74^{\circ}\text{C}$  and the battery MOSFET at  $28^{\circ}\text{C}$ . Due to the highest output voltage value of this module, a higher power can be sunk from our electronic load for the same current and that is why this variant is tested at a higher power level. Again, the thermal camera view indicates a uniform temperature distribution.

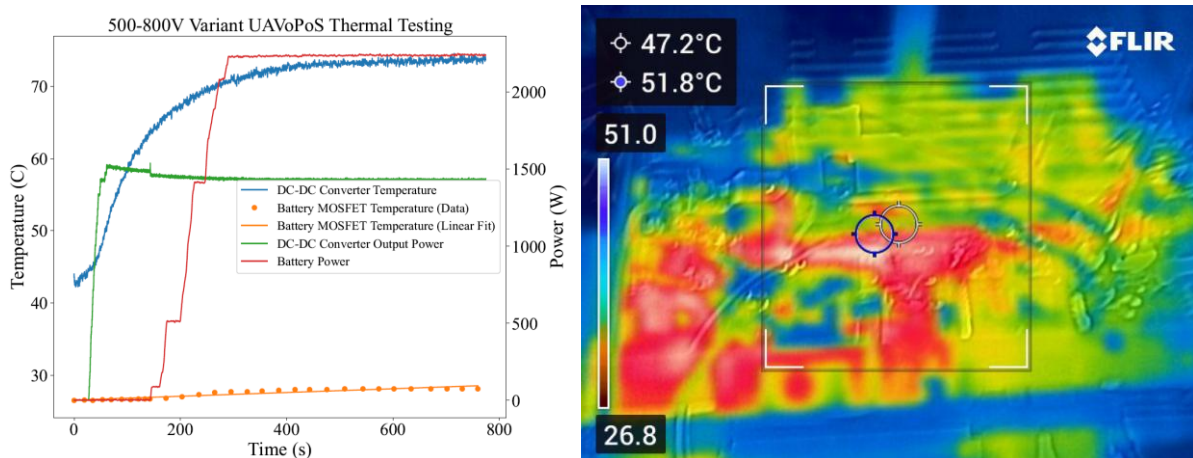


Figure 23: Thermal performance of the 500-800V variant UAVoPoS at full load.